CAIRO: CODING SCHOOL

EF & Linq To Entity

Project

Cairo Coding School



CAIRO: CODING SCHOOL

EF

Code-First Approach



Facebook Project

This Project represents a social media platform similar to Facebook. The system allows users to:

- Create posts
- Comment and reply to comments
- Like posts
- Add friends
- Join groups

CAIRO: CODING SCHOOL

Project Requirements

- The system will be built using OOP principles and the Code-First Approach
  - Create a console app and a class library
  - Separate modals and DTOS

let's think step by step

The Users table is the central entity in the system.

- A user can: Create posts
  - Each post is created by one user.
  - A user can create multiple posts.

let's think step by step

The Users table is the central entity in the system.

- Each user has basic information (Name, Email).

- Post has Likes
  - Users can like posts.
  - A post can be liked by many users.

let's think step by step

The Users table is the central entity in the system.

- Each user has basic information (Name, Email).

- Friendships
  - Users can be friends with other users.
  - Status ??
    - Pending
    - Accepted
    - Rejected

let's think step by step

The Users table is the central entity in the system.
    - Each user has basic information (Name, Email).
- Groups
    - Represents social groups
- GroupMembers
    - Users can join groups.
    - JoinedAt ???

let's think step by step

The Users table is the central entity in the system.

- Each user has basic information (Name, Email).

- UserProfiles
  - Each user has exactly one profile.

let's think step by step

The Users table is the central entity in the system.
    - Each user has basic information (Name, Email).
- Post has Comments
    - Comments belong to a post and are written by a user.

let's think step by step

The Users table is the central entity in the system.
- Each user has basic information (Name, Email).

- Post has Comments
  - Comments can also reply to other comments.
    - Self-reference:
      - A comment can have a parent comment

Project Requirements

⚙️ Features

- User
  - Register with:
    - Username
    - Email
    - Password
    - Bio
    - Date of Birth
  - Login using:
    - Email
    - Password

Project Requirements

⚙️ Features

- User
  - Validation Rules
    - Email must be unique
    - Username must be unique
    - Password minimum length = 8

Project Requirements

⚙️ Features

- Create or update a post
  - Post text must contain more than 5 words and fewer than 100 words
- Delete a post
- Like a Post
- Comment on a Post
- Reply to Comments

Project Requirements

⚙️ Features

- Users can view post details, including:
  - Post creator
  - Creation date
  - Number of likes
  - List of likes
  - Comments
  - Replies

Project Requirements

⚙️ Features

- User can:
  - Send a friend request
  - Accept or Reject a Friend Request
  - Remove friend
  - View friends list
  - View a friend's profile including:
    - Name
    - Email
    - Posts

Project Requirements

⚙️ Features

- Users can go to the home page where they can view:
  - Friends’ posts
  - Group posts
  - Latest posts ordered by date

Project Requirements

⚙️ Features

- User can
  - List all Groups
  - Join a Group
  - View Group Posts
  - Leave a Group
  - Create a group
    - Become the group admin after creating the group
  - Group Admin can:
    - Delete the group
    - Deleting a group will:
      - Delete all posts related to the group
      - Remove all users from the group

Project Requirements

⚙️ Features

- User can
  - Search posts containing the keyword
  - See the common friends
  - Notifications
    - Someone liked your post
    - Someone commented
    - Friend request accepted